Determination of Energy Scale and Resolution of Hadronic Jets Using the Forward Folding Technique

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Abstract

This thesis studies the calibration of hadronic jets using the Forward Folding technique, applied to Monte Carlo (MC) simulated Z+jet events in the ATLAS detector. The analysis aims to determine the mean detector response, a calibration factor, and the detector resolution, defined as the standard deviation of the detector response distribution. These parameters are extracted by comparing reconstructed jet data to truth-level distributions modified using Gaussian smearing, simulating detector effects. The smearing is applied through a convolution process and the optimal parameters are determined using a χ^2 minimization method. In addition, this study deals with challenges during the analysis, such as code inefficiency, statistical fluctuations, and large uncertainties due to a limited dataset at higher energies. These were mitigated by optimizing the code with NumPy vectorization, reducing runtime, and averaging over values from multiple random seeds to reduce statistical fluctuations. The derived calibration and resolution results are consistent with established models, demonstrating the validity of the method in quantifying the detector effects. This study lays the foundation for the forward folding approach to jet calibration with room for improvement through refined error analysis, adaptable modeling across datasets, and future testing with real detector data. A key aspect is that this method makes it possible to determine both energy scale and resolution parameters using a unified method, a novel approach to jet calibration in the ATLAS experiment.

1 Introduction

The core aspect of particle physics is understanding fundamental particles and forces of the Standard Model. The Large Hadron Collider (LHC) at CERN is at the forefront of this scientific endeavor. It provides insights into the interaction between particles and the search for new physics, pushing the boundaries of our understanding and knowledge. The LHC achieves this by conducting high-energy particle collisions and interpreting the resulting experimental data. Large numbers of highly energetic quarks and gluons are created in these high-energy proton-proton collisions at the LHC. These particles quickly undergo an intricate process known as hadronization, producing a collimated spray of hadrons. These sprays form a distinct cone shape and are known as jets. Jets are the most frequently observed phenomena in the ATLAS experiment, constituting a significant portion of the visible particle content at the LHC. Thus, accurately measuring and understanding their properties is essential [6].

However, accurately measuring jets is challenging as the data from the reconstructed jets in a detector have been impacted by pile-up from previous collisions, detector effects, resolution imperfections and more. These lead to measurement uncertainties in the reconstruction of the actual collision and jet data. Meticulous analysis techniques are used to address statistical and systematic uncertainties. These discrepancies must be carefully corrected by applying jet calibration [3]. One of the stages of jet calibration is the use of Z bosons as reference objects. Z bosons are ideal references as they are well understood and measured with high precision. This is because jet calibrations depend on independently measuring and calibrating the energy of photons or the lepton decay products of a Z boson, specifically through the lepton decay channels ($Z \to e^+e^-$ and $Z \to \mu^+\mu^-$). Therefore, this technique can be used to derive the jet response of the ATLAS Z+jets data [12].

This thesis explores a novel technique called 'Forward Folding' to measure the ATLAS detector's response to jets. Traditional methods work by removing the detector effects and imperfections by unfolding measured data. In contrast, this method extracts the mean jet response and the detector resolution of jets by modifying simulated data, which depicts a perfect detector's observations. This simulated data is shifted and smeared to best match the actual data collected by the detector [8]. The principal mathematical concept behind the forward folding technique is convolution. The project aims to better understand jet measurements and study this method with ATLAS Z+jets [15].

2 Theoretical Background

2.1 Jets in Particle Physics

In particle colliders, high-energy particle collisions lead to jets due to quarks and gluons produced during proton-proton collisions. However, quarks and gluons cannot exist freely in nature due to color confinement [11]. Quarks are held together by gluons, which have a constant energy per unit length. A quark-antiquark pair flying apart would stretch the gluon, increasing energy. Once the energy is high enough, the gluon can snap and produce a quark-antiquark pair from the energy it contains. This process is repeated multiple times [16], and finally, these quarks and anti-quarks form hadrons together since they cannot exist freely. This hadronization process creates pions, kaons, and baryons, resulting in a spray of hadrons known as a jet [11].

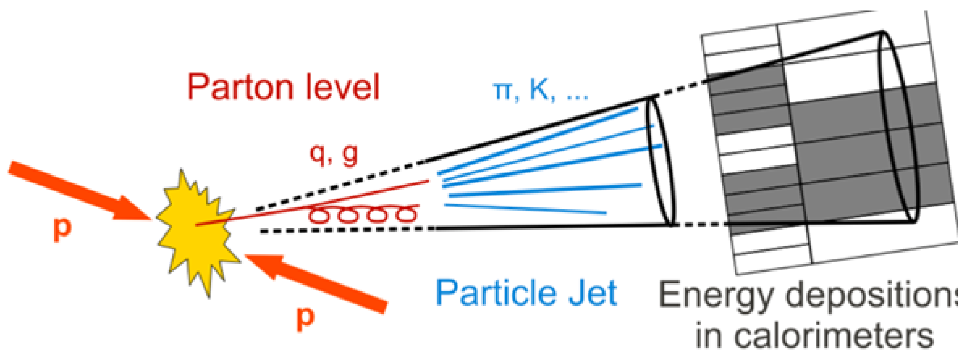

Figure 1: Creation of a jet, transition from parton-level (quarks, gluons) to final-state particles (e.g., π, K) Figure from ref. [11].

2.2 Jet Measurement Challenges

The measurement of jet properties proves to be a challenge in high luminosity environments, primarily due to detector effects such as pile-up and detector resolution. Pile-up impacts jet measurements because of additional proton-proton collisions create overlapping jets in the same area as the jet event in focus. These overlapping jets introduce extra energy and particles unrelated to the jet of interest. Furthermore, pile-up from energy deposits from other collisions creates remnants in detectors, increasing the fluctuations in background noise and artificially inflating the jet energy. These effects impact the accuracy and precision of jet measurements [1].

Detector resolution is the measure of the ability to reconstruct the jet and its properties precisely. Due to detector limitations, jet measurement fluctuations introduce uncertainty known as Jet Energy Resolution (JER). The JER measures the precision of the detector in its ability to distinguish between jets of similar energies. This could be evaluated by simulating dijet events with known jet properties and analyzing the spread in the reconstruction of the jet [18].

The JER is important for accurately reconstructing the jet and its properties. The relative JER as a function of the transverse momentum of the jet can be parameterized using a functional form based on calorimeter resolutions. The function incorporates three terms: noise (N), stochastic (S) and constant (C) [9]:

$$\frac{\sigma(p_T)}{p_T} = \frac{N}{p_T} \oplus \frac{S}{\sqrt{p_T}} \oplus C. \tag{1}$$

The noise (N) term arises from electronic noise and pile-up, which is expected to be significant in the low transverse momentum ranges. The stochastic (S) term accounts for the statistical fluctuations in energy deposition, which dominates the resolution in the intermediate transverse momentum ranges. Lastly, the constant (C) term represents the energy loss due to instrument effects and the non-uniform detector response, which is prominent in high transverse momentum ranges [9].

In addition to the JER, the Jet Energy Scale (JES) is a critical source of systematic uncertainty in the measurement of jets, quantifying the discrepancy between true jet energy properties and the reconstructed jet energy properties. This energy shift is due to the hadron calorimeters not exhibiting a linear or uniform response to the jets. Doubling the jet energy might not double the detector response, and different parts of the detector may respond differently to the same energy jet. This introduces uncertainties and biases in the measured jet energies [17].

These challenges with pile-up effects, detector resolution, and JES uncertainties are intricately linked and collectively impact the reliability of the reconstructed jets' measurements. To address these issues, the forward folding jet calibration technique is employed in this thesis. Understanding and correcting these systematic errors will improve the quality of jet measurements with improved accuracy and precision. In order to quantify these challenges, the momentum of the jets must be precisely determined. This requires the jets to be measured against a reference with well-known momentum.

2.3 Z+Jet Events

At the Large Hadron Collider, Z bosons are produced along with jets that originate from quark-gluon, quark-antiquark, or gluon-gluon interactions during proton-proton collisions. These events are known as Z+jets, important for jet calibration and precise mass measurements [23]. The Drell-Yan process is where the Z boson decays into a lepton-antilepton pair, specifically electrons $Z \to \mathrm{e}^+\mathrm{e}^-$ or muons $Z \to \mu^+\mu^-$. This process is measured with exceptional precision, making them a standard candle process for detector calibration [21, 4]. The observable in interest from this process is the transverse momentum of the Dilepton process, which is accurately measured and serves as a reliable and ideal reference.

This is ideally suited for jet calibration as the transverse momentum of the reconstructed jet should balance that of the reference Z jet transverse momentum in back-to-back events where the momentum should be equal [12]. This would allow us to quantify the detector effects on the jet measurements. Z+jets will prove to be useful in determining important parameters such as the Jet Energy Scale (JES) and the Jet Energy Resolution (JER). The approach in this thesis is to use the Z+jets to implement the Forward Folding method for jet calibration. The comparison between the reconstructed jet momentum and the precise Dilepton system momentum will aid in the correction of systematic errors and improve accuracy in jet measurements.

2.4 Convolution and Forward Folding

The approach of this thesis to extract key parameters, such as the Jet Energy Scale (JES) and Jet Energy Resolution (JER), is to simulate the detector effects and apply the distortions to an idealized truth distribution, which will then be compared directly with the experimentally measured distribution. Mathematically, this process is known as convolution, a mathematical operation that 'blends' two functions by analyzing how one function g overlaps with another function f as it shifts across it. The values of the two functions f and g are multiplied and then summed together to produce a value. This process will repeat over the chosen range and generate a new function that is the convolution of f and g. It is mathematically defined as [25]:

$$(f * g)(t) \equiv \int_0^t f(\tau)\, g(t-\tau)\, d\tau \tag{2}$$

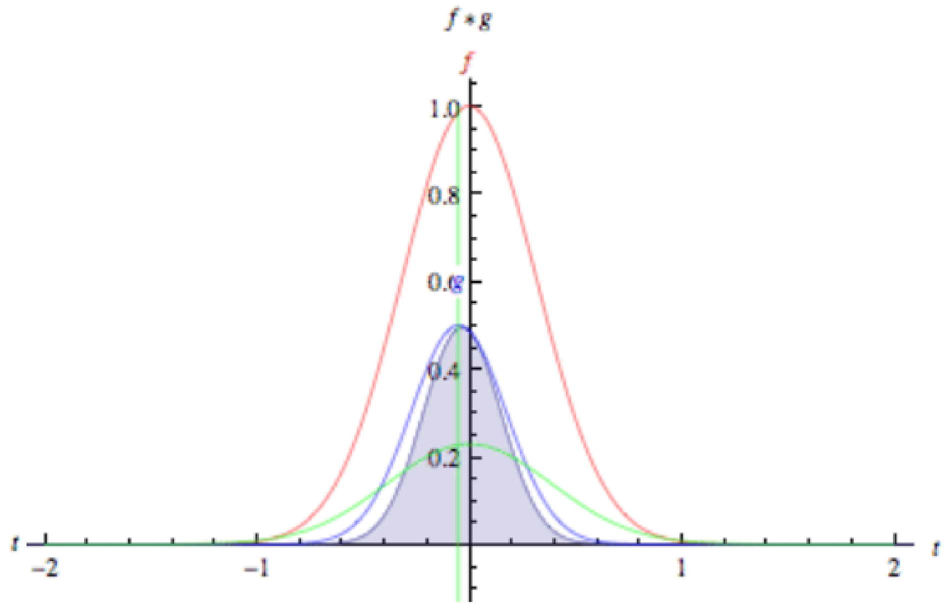

Figure 2: Illustration of convolution. Figure from ref. [25]

Where f (red curve) represents the true distribution and g (blue curve) is the function that simulates the effects of the detector. The green curve is the convolution of both functions, the folded distribution [25].

The convolution comprises the detector effects and yields a smeared distribution. The process and method used in this thesis is Forward Folding, which is a convolution of the theoretical distribution with a detector effects function, in this case, Gaussian smearing. This results in a 'folded' distribution which is shifted and stretched and be compared directly to the experimentally measured distribution. This is in contrast to the unfolding procedure, which aims to recover the truth distribution from the measured data by removing detector effects [13].

The Gaussian function used is given by:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{3}$$

Where, the μ represents the is the mean shift, and σ is the smearing width, representing the JES and JER respectively.

This thesis has employed Forward Folding as the principal method to calibrate the ATLAS detectors' response to jets. The parameters JES and JER can be extracted by comparing convoluted truth distributions to measured distributions. This is achieved using chi-square minimization. The optimal values of mean and sigma values that minimize chi-square would be the best estimates of JES and JER respectively [8].

3 Methodology and Approach

This section presents the results and analysis of the Forward Folding Technique to calibrate jets measured in the ATLAS detector. This method is applied to carefully selected events, filtered out based on well-defined jet observables. Gaussian smearing is then applied and statistically optimized to yield the jet energy resolution of the detector.

3.1 Data and Key Observables

The application of Forward Folding begins with a dataset consisting of Z+jet events. The analysis in this thesis is based on proton-proton collisions by the ATLAS experiment at the Large Hadron Collider (LHC). This analysis uses Monte-Carlo (MC) sample datasets, containing both reconstructed and truth-level data, which are ideal for testing new calibration methods and techniques and serving as a reference for calibration. These MC sample datasets were generated using the Powheg event generator [19, 5], interfaced with Pythia8, which simulates parton showering and the hadronization process [22]. This process generates the truth-level particles and then is processed through a very advanced simulation model based on Geant4, a simulation model. This simulation accounts for every aspect of the ATLAS detector, particle interactions, energy depositions and more, representing what a perfect detector would see in the absence of detector effects. The model could also simulate detector effects like pile-up collisions and resolution effects, producing measurements similar to real detector data [7]. The upside to simulated data is the access to truth data (Ideal distribution) and detector data (Measured distribution). This analysis uses the simulated data to test the method and approach to quantify the JES and JER effects.

Each Z+jet event has key observables:

- p_{T}^{j1} - the reconstructed jet transverse momentum, comprising of detector effects
- $p_{\mathrm{T}}^{\ell\ell}$ - the transverse momentum of the Z boson (dilepton system as $Z \to \ell\ell$),
- $p_{\mathrm{T}}^{\mathrm{truth},j1}$ - the truth jet transverse momentum (what a perfect detector would see)

In this technique, the transverse momentum of the jet p_{T}^{j1} is the main observable of interest. As previously mentioned, jets measured by the detector are subject to detector effects and experimental and instrumental conditions, leading to systematic uncertainties in the measured values. To accurately assess and correct these systematic uncertainties, the transverse momentum of the Z boson $p_{\mathrm{T}}^{\ell\ell}$ is used as a reference. The Dilepton system $Z \to \ell\ell$ is well understood and measured with high experimental precision, making it an ideal reference for calibration [21].

From the above observables we define the direct balance quantity as the ratio of jet p_{T} to dilepton p_{T}:

$$r_{\mathrm{DB}} = \frac{p_{\mathrm{T}}^{j1}}{p_{\mathrm{T}}^{\ell\ell}}$$

The forward folding method fundamentally relies on this observable. This ratio reflects the extent of detector effects on the jet measurements. In a perfect detector, the ratio is $r_{\mathrm{DB}} = 1$, meaning that the momentum of the jet and the Z boson is balanced. The distribution of this ratio will reveal the impact of the detector effects. If $r_{\mathrm{DB}} < 1$, the jet energy is underestimated, and if $r_{\mathrm{DB}} > 1$, the jet energy is overestimated. Two key calibration parameters can be extracted from this distribution, the mean jet detector response and the jet resolution. The Jet response is the magnitude of the mean shift in the r_{DB} distribution, while the Jet resolution is the standard deviation (width) of the r_{DB} distribution. These parameters represent systematic errors and measurement uncertainties [12, 8].

Monte Carlo simulations provide truth jet data ($p_{\mathrm{T}}^{\mathrm{truth},j1}$), representing what a perfect detector would see. The forward folding technique is then applied to the truth distribution using a Gaussian smearing function. The Gaussian smearing function broadens and shifts the truth distribution, simulating detector resolution and response. The smeared truth distribution is compared to the r_{DB} distribution. The jet resolution will be precisely determined through iterative adjustments by varying the smearing parameter sigma (σ).

3.2 Event Selection Criteria

To ensure an accurate and reliable jet calibration, events were filtered based on a specific criterion to minimize background noise such as pile-up effects. The selection criteria is:

- The jet transverse momentum p_{T}^{jet} is constrained to a chosen range (Ex. $100 \leq p_{\mathrm{T}}^{jet} < 140$ GeV). To limit the analysis to a well-defined range of momentum and avoid uncertainties from wide momentum distributions.
- The jet rapidity, which is the angle of the jet from the beam axis [14], $|y_j| < 0.7$, to focus on the central detector regions where the measurements are the most accurate and precise [2].

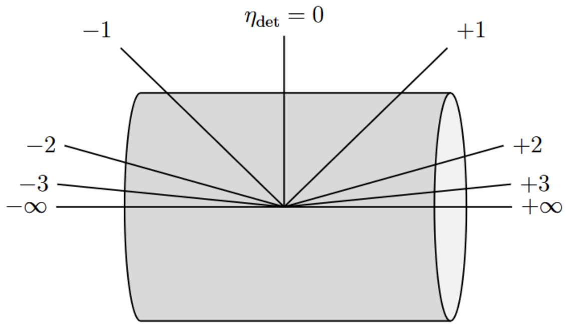

Figure 3: Detector pseudorapidity η, which describes the angular distribution of particles relative to the beam axis. Figure from ref. [20].

- The angular separation, which is the azimuthal angle, requires to be $\Delta\phi > 2.9$ radians, ensuring the coverage of near back-to-back events for the dilepton system to balance the jet in the azimuthal plane [12].

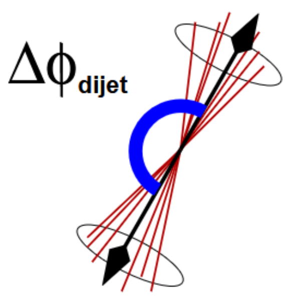

Figure 4: Azimuthal angle $\Delta\phi$ between two jets in a back-to-back dijet event. Figure from ref. [10].

The implementation of these criteria is as follows:

```

def extracting_data(tree, min_pT, max_pT):
    truth_pT_j1_data = []
    tree_pT_ll_data = []
    tree_pT_j1_data = []

    # Looping over all events in the ROOT file
    for i in range(tree.GetEntries()):
        tree.GetEntry(i)

        # Selection 1: Checking if pT_ref is within chosen range
        if tree.pT_ref < min_pT or tree.pT_ref > max_pT:
            continue

        # Selection 2: Jet Rapidity checked  to be within central region
```

```python
        if abs(tree.yDet_j1) > 0.7:
            continue

          #
        if tree.truth_ll_match == 0 or tree.truth_j1_DR > 0.2:
            continue
        # Selection 3: Checking if jets are nearly back-to-back
        if tree.Dphi_ll_j1 < 2.9:
            continue

        # Data from events passing the selection criteria stored in datasets
        truth_pT_j1_data.append(tree.truth_pT_j1)
        tree_pT_ll_data.append(tree.pT_ll)
        tree_pT_j1_data.append(tree.pT_j1)


    truth_pT_j1_data = np.array(truth_pT_j1_data)
    tree_pT_ll_data = np.array(tree_pT_ll_data)
    tree_pT_j1_data = np.array(tree_pT_j1_data)

    # Returning the extracted relevant data
    return truth_pT_j1_data,tree_pT_ll_data,tree_pT_j1_data
```

Listing 1: Python code implementing selection criteria

The full selection criteria can be summarized as:

$$\begin{aligned}
&p_T^{\text{jet}} \in [p_T^{\text{min}},\ p_T^{\text{max}}],\\
&|y_j| < 0.7,\\
&\Delta\phi_{\ell\ell} > 2.9,
\end{aligned}$$

3.3 Histogram Construction

The direct balance ratio r_{DB} is computed for the selected jet events to study the impact of detector effects on jet measurements. These values are then used to construct histograms to visualize the jet measurements relative to the well-measured dilepton system. An example comparison between the truth-level and reconstructed distributions in the 50-100 GeV momentum range is shown below. The visual shift and widening highlight the discrepancy caused by the detector effects. A chi-square value was calculated to quantify the discrepancy.

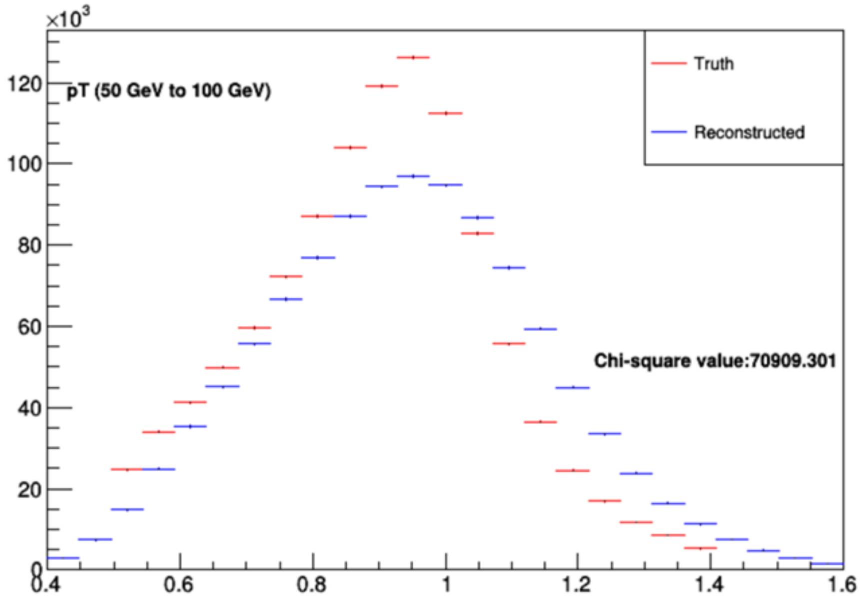

Figure 5: Comparison of the truth-level (red) and reconstructed (blue) $r_{\mathrm{DB}} = \frac{p_T^{j1}}{p_T^{\ell\ell}}$ distributions in the $50\,\mathrm{GeV} < p_T^{\mathrm{ref}} < 100\,\mathrm{GeV}$ range. The x-axis shows the direct balance ratio, while the y-axis indicates the number of events per bin. The χ^2 value quantifies the level of agreement between the two distributions.

In this histogram, the mean of the distribution is shifted slightly to the right, indicating that the jet energy scale (JES) implies an overestimation of the jet energy. Additionally, the standard deviation (width) of the distribution corresponds to the jet energy resolution (JER), broadened due to resolution effects of the detector.

Ideally, a well-matched histogram is where the convoluted truth histogram with the applied detector effects aligns closely with the measured distribution. This would mean that the parameters used for the jet calibration models the detector effects.

3.4 Gaussian Smearing and χ^2 Minimization

To simulate detector effects, a Gaussian smearing function was applied to the truth-level distribution. This function introduces a blur to each truth value representing the inaccuracies introduced in the detector. The smearing values are randomly generated from a standard Normal (Gaussian) Distribution with the given mean and sigma, quantifying the JES and JER, respectively.

In this phase of the analysis, μ is fixed to 1 to isolate the resolution effects of the detector. The focus on determining the optimal σ value first will help validate the method before making it more complex. Once optimal σ values for each energy range are identified, the JER curve curve can be constructed as a function of jet energy.

To identify the optimal smearing width σ, a broad scan over a range of σ values is done. Smearing was applied to the truth-level data for each σ value. A histogram of the convoluted truth distribution was created using the smeared values. The convoluted truth histogram is compared to the measured reconstructed histogram using a χ^2 test. The σ value with the minimum χ^2 was considered the best estimate.

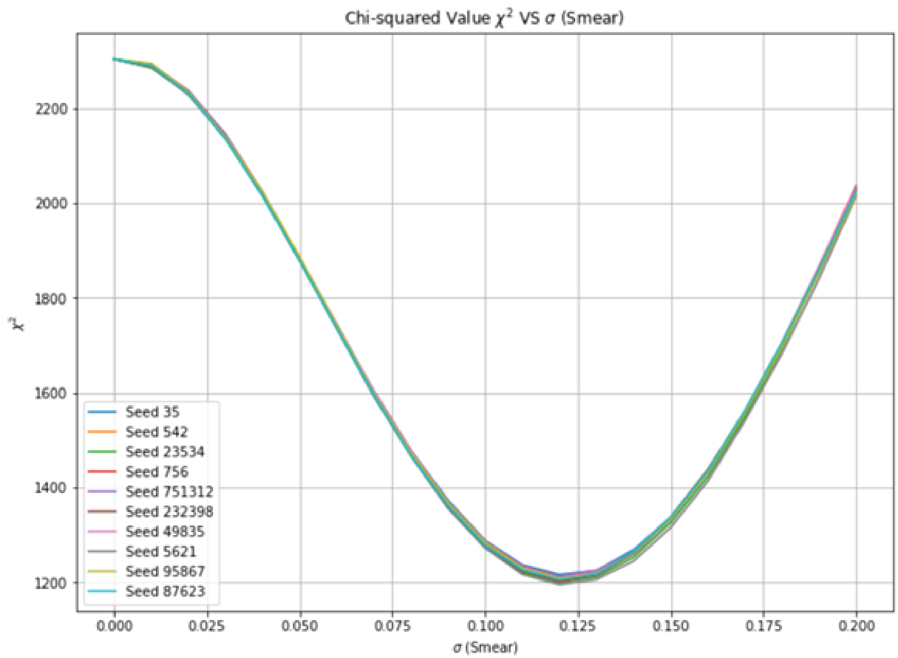

Figure 6: Chi-square values χ^2 as a function of the smearing parameter σ for 10 different random seeds.

Following that, a finer scan of 100 σ values in a narrower interval about σ values corresponding to χ^2 values within 2% of the maximum χ^2 value found in the first scan. This two-step scan process is repeated for 10 random seeds. This process makes the method more efficient and precise, avoiding unnecessary data and minimizing statistical noise.

The χ^2 values of the 10 seeds are then averaged to account for statistical fluctuations, yielding an averaged χ^2 vs σ curve. Since the shape of the curve resembles a parabola near the minima, a quadratic function was fit to the data points to further smoothen out the fluctuations. The corresponding σ value associated with minimum χ^2 from this quadratic fit was taken to be the estimate of the optimal resolution parameter σ_{optimal}.

The Quadratic Fit used is given by:

$$\chi^2(\sigma) = a\sigma^2 + b\sigma + c \tag{4}$$

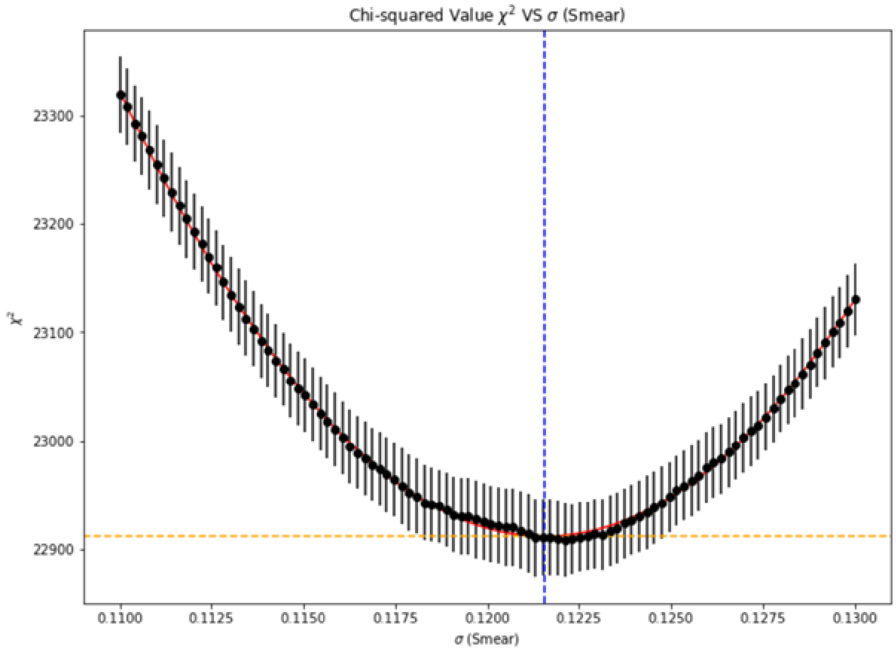

Figure 7: Chi-square values (χ^2) as a function of the smearing parameter σ for a single jet p_T bin.

Each black point represents the mean χ^2 computed over multiple random seeds, with error bars denoting statistical uncertainty. The orange dashed line shows the minimum χ^2 value, and the vertical blue dashed line marks the corresponding optimal σ value. The parabolic shape near the minimum justifies the use of a quadratic fit.

The error on the σ_{optimal} value was determined using the well-known Chi-Square confidence interval method. This analysis determined the jet resolution parameter for an energy range. This approach ensures the precise application of the forward folding calibration.

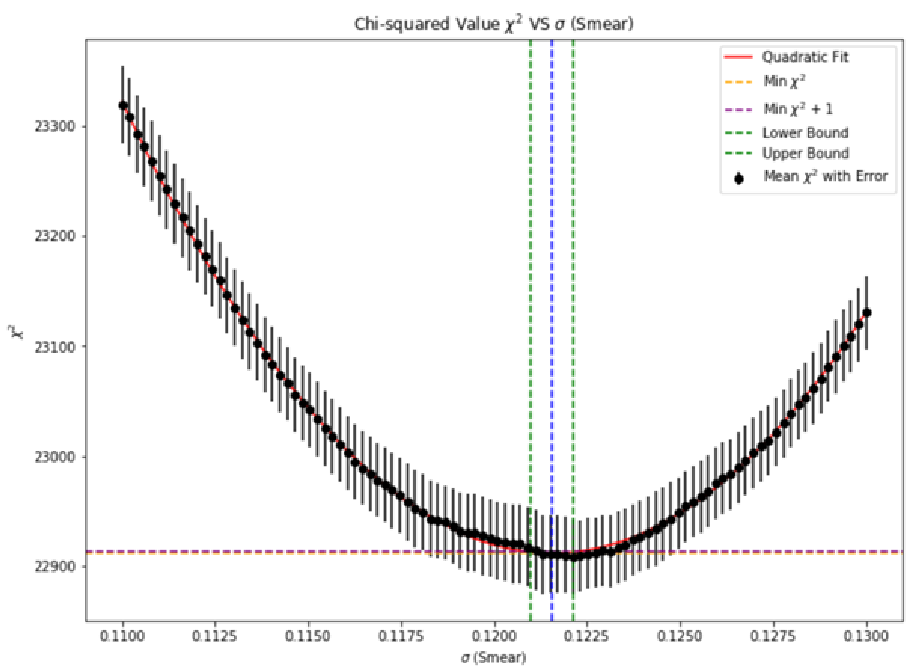

Figure 8: Quadratic fit (red curve) to the mean χ^2 values for different smearing parameters σ.

The optimal σ is indicated by the blue vertical dashed line, while the green dashed lines represent the upper and lower bounds defined by $\chi^2 = \chi^2_{\text{min}} + 1$. The half-width corresponds to the uncertainty on the estimated jet resolution parameter. The optimal σ computed for the $50\,\text{GeV} < p_T^{\text{ref}} < 100\,\text{GeV}$ range was 0.122.

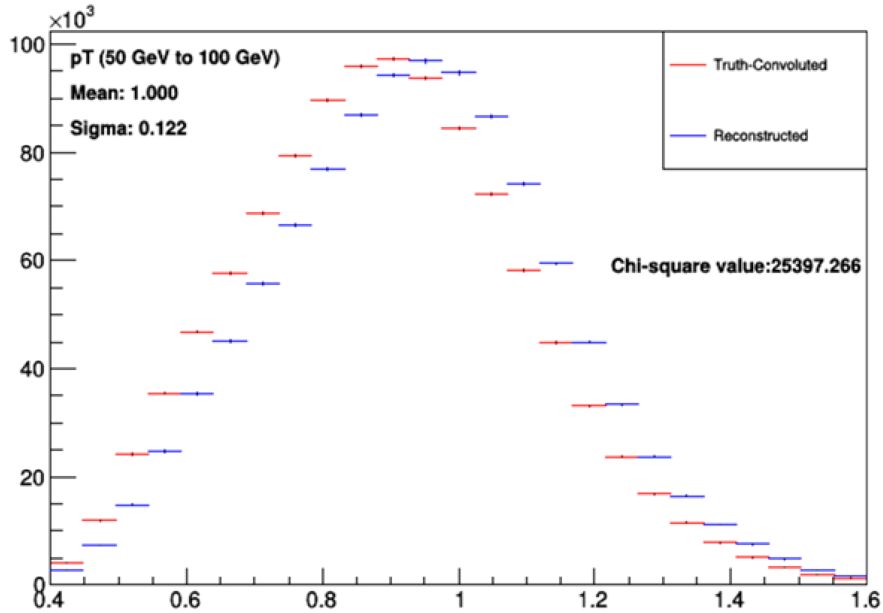

Figure 9: Comparison after applying the optimal smearing parameter ($\sigma = 0.122$) to the truth-level distribution.

To demonstrate the effectiveness of this smearing parameter, the optimal sigma was applied to the truth-level distribution. Figure 5 shows the comparison between the initial truth-level histogram and the reconstructed histogram before smearing. A large chi-square value (70909) quantifies the discrepancy between the two histograms. Figure 9 above shows the result after the Gaussian smearing using the optimal sigma value. The agreement between the histograms has improved significantly, visually matching better in both peak height and width of the distribution. The chi-square value has decreased by over 60%, validating the sigma estimate derived through the forward folding technique. With the optimal sigma for the given jet range is determined, the smearing width is fixed. The process is repeated with varying the mean values, to derive the Jet Energy Scale (JES) parameter, the optimal mean. The optimal μ computed for the $50\,\text{GeV} < p_T^{\text{ref}} < 100\,\text{GeV}$ range was 1.054.

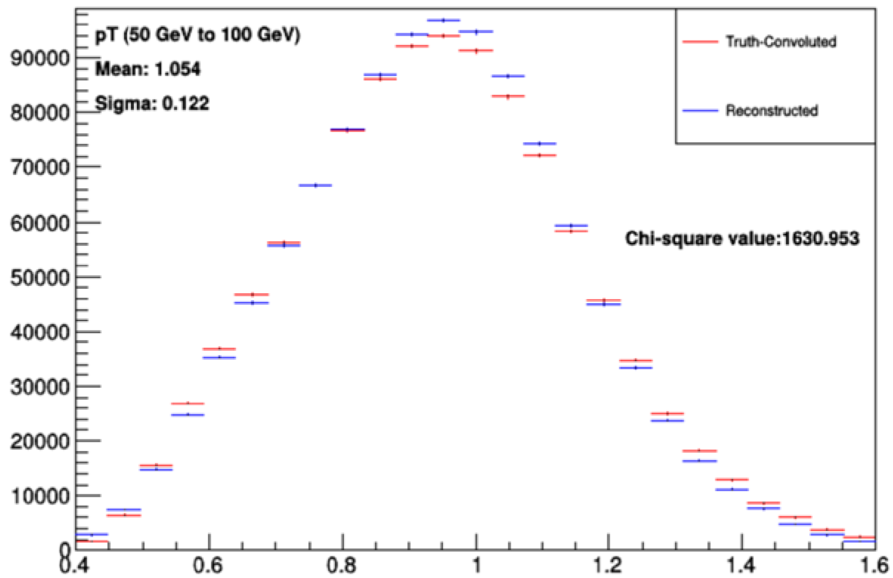

Figure 10: Comparison between reconstructed data and truth-convoluted histogram using the fully optimized smearing parameters: $\mu = 1.054$, $\sigma = 0.122$.

Figure 10 shows the improved alignment and low chi-square value(1631), around a 97% reduction. This confirms the successful calibration of the resolution (JER) and scale (JES) parameters, effectively modeling the detector effects. This two-stage approach, by first deriving the JER parameter and then using it to identify the JES parameter, allows a more precise and independent calibration of the jet measurements. This approach is applied across a range of p_T values to construct the full JES and JER curves.

4 Results

4.1 Jet Energy Resolution

The optimal sigma values for each p_T range are presented in the table below. These values represent the jet energy resolution, the standard deviation of the r_{DB} distribution. The corresponding uncertainties were extracted using the chi-square confidence interval method.

p_T^{ref} Range (GeV)	JER (σ)	σ_{J}ER
50 – 60	0.1261	0.0040
60 – 72	0.1204	0.0041
72 – 86	0.1149	0.0044
86 – 104	0.1044	0.0051
104 – 125	0.0970	0.0060
125 – 150	0.0922	0.0071
150 – 180	0.0867	0.0084
180 – 216	0.0827	0.0109
216 – 260	0.0737	0.0155
260 – 310	0.0739	0.0212
310 – 375	0.0681	0.0311
375 – 450	0.0634	0.0440
450 – 540	0.0721	0.0577

Table 1: JER dataset: Jet energy resolution (JER) values and corresponding uncertainties for each jet p_T range

The table above clearly shows that the resolution improves as the jet energies increase, by having lower JER values. This behavior is consistent with established results.

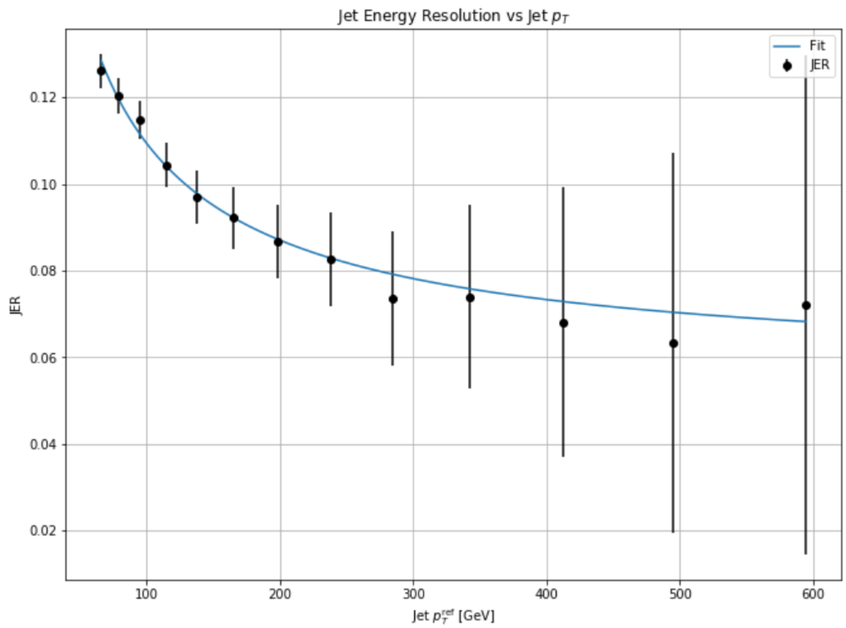

Figure 11: Jet Energy Resolution (JER) as a function of jet p_T^{ref}.

The black points represent the extracted JER values for different p_T ranges, with the error bars indicating the statistical uncertainties. The table and figure above clearly show that the resolution improves with increasing jet p_T. The extracted JER values show a decreasing trend with increasing jet p_T, consistent with established results presented in Ref [9].

The blue curve shows the fitted function to the JER curve:

$$\frac{\sigma(p_T)}{p_T} = \sqrt{\left(\frac{N}{p_T}\right)^2 + \left(\frac{S}{\sqrt{p_T}}\right)^2 + C^2}$$

The fitted parameters were: $N = 0.0000 \pm 0.0000$, $S = 0.9381 \pm 0.0574$, and $C = 0.0564 \pm 0.0097$. The Noise term (N) was found to be negligible, the stochastic term dominates, stating the presence of statistical fluctuations in energy deposition affecting resolution, and the constant term C reflecting the calibration uncertainties due to technical effects.

4.2 Jet Energy Scale

The optimal mean values for each p_T range represent the Jet Energy Scale (JES), the mean change in the detector's measurement of the jet energy compared to the truth-level distribution. The extracted JES values and their associated uncertainties are presented in the table below.

p_T^{ref} Range (GeV)	JES (μ)	σ_{JES}
50 – 60	1.0579	0.0022
60 – 72	1.0515	0.0024
72 – 86	1.0464	0.0027
86 – 104	1.0393	0.0031
104 – 125	1.0334	0.0037
125 – 150	1.0287	0.0044
150 – 180	1.0247	0.0055
180 – 216	1.0225	0.0071
216 – 260	1.0190	0.0094
260 – 310	1.0207	0.0122
310 – 375	1.0219	0.0180
375 – 450	1.0178	0.0256
450 – 540	1.0236	0.0359

Table 2: JES dataset: Jet Energy Scale (JES) values and corresponding uncertainties for each jet p_T range

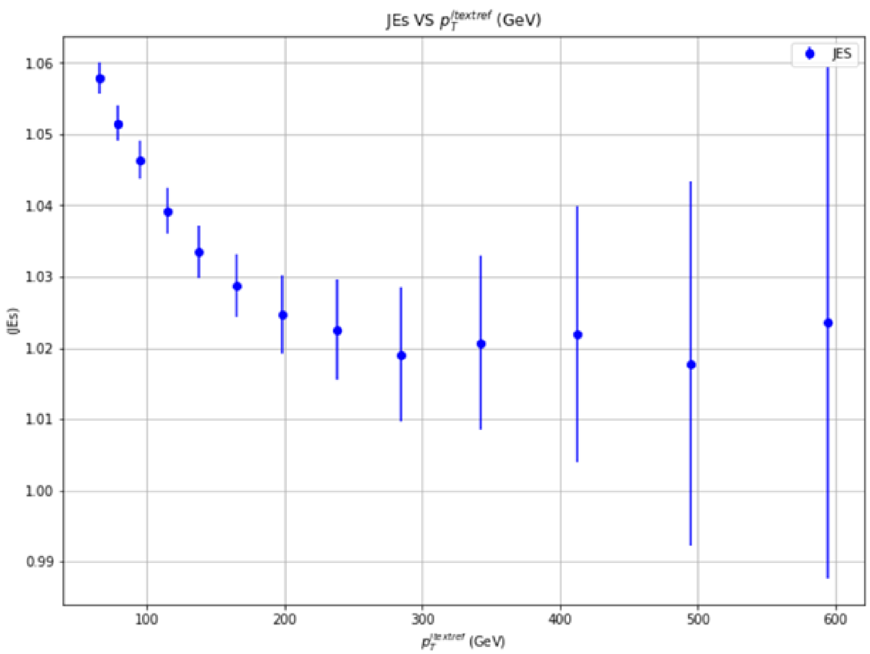

Figure 12: Jet Energy Scale (JES) as a function of jet p_T^{ref}.

The blue points represent the extracted JES values for different p_T ranges, with the error bars indicating the statistical uncertainties. The JES curve quantifies the systematic shift in the reconstructed jet energy. The table and figure above clearly show that the systematic bias improves with increasing jet p_T. The upward trend in the low p_T energy range reflects the overestimation by the detector, which flattens out at higher energies, indicating a reduction in detector bias in the jet measurements, which is consistent with the resolution behavior.

5 Discussion

This study successfully demonstrated the application of the Forward Folding technique to determine the Jet Energy Scale (JES) and Jet Energy Resolution (JER) using MC Z+jet data. The primary objective of this study was to understand jet measurements by applying jet calibration modeling to account for detector effects. The results agreed with the established models to a reasonable extent, supporting the method and approach. However, the significant uncertainties at higher jet energies question the precision of the parameters needing refinement of the method.

Initially, early versions of histogram convolution and χ^2 comparisons were computationally intensive. The processing time for a single p_T range would take hours, making testing and debugging inefficient for large datasets. From a coding standpoint, the bottleneck was iterating over every single value, one by one, using three nested loops. This was resolved by shifting from an iterative process to a NumPy based approach using vectorized operations, where computations are made on the entire array of values simultaneously. This approach significantly reduced the runtime of the code, allowing more efficient analysis over different seeds and p_T ranges. This optimization was critical for the analysis of large datasets.

Following that, the built-in χ^2 test in ROOT raised warnings stating low bin content, which affected the precision of calculations. Therefore, to gain a better understanding, a manual χ^2 method was implemented to work around the issue.
The χ^2 implemented is defined by:

$$\chi^2 = \sum_{i=1}^{N_{\text{bins}}} \frac{(n_{c,i} - n_{r,i})^2}{\epsilon_{c,i}^2 + \epsilon_{r,i}^2}$$

For each histogram bin i, $n_{r,i}$ is the bin content from the reconstructed histogram, $n_{c,i}$ is the bin content from the truth-convoluted histogram, and the respective uncertainties for each $\epsilon_{r,i}$ and $\epsilon_{c,i}$. This approach gave more understanding of the method. Along with this, some p_T ranges had too few events to yield reasonable results. Thus, p_T ranges with less than 100 events were omitted.

Next, the χ^2 vs σ plots showed statistical fluctuations due to random smearing. To ensure statistical stability, the χ^2 minimization was repeated for ten different seeds and the results were averaged. The resulting curve with lower fluctuations was then fit with a quadratic function to locate the minimum more precisely. Following that, a fixed χ^2 threshold was used to find the zoomed-in region for further analysis. This threshold did not work for all p_T ranges as the number of events and noise were varied. Rather than using a fixed chi-square threshold, a dynamic threshold was set as a percentage of the maximum observed χ^2 value in each range. This ensured that a reasonable zoom-in window could be analyzed for any range of event counts or chi-square values.

After the data analysis, a major observation from the JER and JES curves was the presence of large error bars, especially at high p_T ranges. These uncertainties are due to the lower number of events at higher energies affecting the precision of the fit. Furthermore, a more refined and adaptive error propagation process might be needed to process the uncertainty of p_T ranges with varying event counts and χ^2 values. Lastly, the fitting function for the JER curve initially yielded negative values and large uncertainties. This issue was resolved by applying parameter constraints in the fitting process to obtain meaningful values.

This study lays a foundation for the Forward Folding method in Jet calibration, with potential to improve. Exploring more smearing models like the double-sided Crystal Ball function to adapt to distributions as a better fit [24]. In the event of available computational power, the best estimate for the JES and JER can be simultaneously obtained. Additionally, a dynamic error analysis is needed to adapt to any event statistics. For future studies, the MC reconstructed data could be replaced with real collision data, extracting genuine JES and JER values. This would allow the forward folding technique to be applied practically. A comparison of results between the MC data and real data would provide the accuracy of the simulation in detector effects and validate the forward folding method. It could also provide valuable insights on refining and improving the current method.

In conclusion, this study effectively applied the Forward Folding technique to determine key detector response parameters, jet energy resolution (JER) and jet energy scale (JES). Using Z+jet simulations, the method was structured to model detector effects and analyze jet distributions. The improvements will help yield a more precise method and adaptable approach to understanding and correcting the detector response, which would increase the applicability of this method in future ATLAS experiments.

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